Gideon Kofi Amo Appau

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Research interests: continual and long-horizon reinforcement learning; world models and model-based RL; learning under data scarcity.

Education

University of Science and Technology of China (USTC) Suzhou, China
M.S. in Software Engineering (AI/ML concentration) *Sep. 2024 – Jul. 2027 (Expected)*

- Thesis (working title): *Class-Adaptive Sharpness-Aware Minimization for Long-Tailed Visual Recognition.* Advisor: Assoc. Res. Pengkun Wang.
- Coursework: Artificial Intelligence, Optimization Theory, Information Theory and Coding.

University of Education, Winneba Ghana
B.Sc. in Information Technology Education; CGPA 3.61/4.0 *Aug. 2018 – Oct. 2022*

- First Class Honours, Top 10%. Advisor: Dr. William Asiedu.

Research Experience

University of Science and Technology of China Suzhou, China
Researcher (M.S. thesis); Advisor: Assoc. Res. Pengkun Wang *2025 – Present*

- Proposed **CA-Focal-SAM**, a class-adaptive sharpness-aware minimization optimizer; improves balanced accuracy over Focal-SAM on CIFAR-10-LT (IR=200, 3 seeds) with no architectural change and zero inference overhead.
- Mechanism study on CIFAR-100-LT (IR 50/100/200; CE, Logit Adjustment, LDAM) isolated a frozen log-scarcity prior as the component carrying the gain: +2.50 pp tail accuracy over Focal-SAM at IR=100 and +1.98 pp at IR=200 (4 fixed seeds each).
- Built **SFCA**, an online controller reallocating SAM regularization from live geometric feedback, and falsified it under a pre-specified protocol (polarity under two outer losses, signal-destruction controls, hindsight-oracle bound, class-level regression): destroyed feedback matched intact feedback, and the oracle lost to the frozen prior - a rigorous negative boundary result.
- Ran a reproducible multi-seed pipeline (single launcher, uniform recipe, fixed seeds; ∼200 training runs) with per-run JSON and per-class CSV records; ImageNet-LT and Places-LT evaluations in progress.

Selected Projects

RL Research Portfolio (github.com/Amo-Gideon/rl-portfolio) *2026 – Present*

- **World model + planning:** on a PyBullet 2-link robot arm, a learned world model (MLP ensemble) with CEM planning reaches 100% single-target success vs. 82% for PPO, and holds 100% under out-of-distribution physics shifts (doubled link mass, 1.5× gravity, halved torque limits) that reduce PPO to 70–80%.
- **Obstacle avoidance:** obstacle clearance as a CEM planning cost cut collision rate from 57% to 23%, where five PPO training configurations failed; also reproduced Dueling/Double DQN from scratch and built an RLHF pipeline (SFT, reward model, PPO) for a 0.5B-parameter language model.

Industry Experience

Future Interns Remote
Machine Learning Intern *Mar. 2026 – Apr. 2026*

- Built a 6-month sales forecasting system and a BERT/DistilBERT ticket-classification pipeline with sentiment-based priority routing.

Akenten Appiah Menkah University Kumasi, Ghana
IT Assistant *Nov. 2022 – Aug. 2023*

- Automated data workflows in Python, cutting manual processing time by 40%.

Amidarr A-mole Ghana
Full-Stack Web Developer Intern *Aug. 2021 – Mar. 2022*

- Built 3 full-stack web applications (Flask, JavaScript, MySQL), integrating the PayStack payment API into an e-commerce platform.

Teaching, Mentorship, and Presentations

Peer Mentor, USTC Software Engineering: guided junior master's students on PyTorch and SAM *2024 – 2025*
Talk: USTC Thesis Progress Seminar, *CA-Focal-SAM for Long-Tailed Visual Recognition* *2026*
Talk: USTC Engineering Practice Defense, *Homomorphic Encryption Deployment on FPGA* *2025*

Honors

Chinese Government Scholarship (CSC), USTC *2024*
First Class Honours, Top 10%, University of Education Winneba *2022*

Skills

Languages:	Python, JavaScript, SQL, Java, Bash
RL:	PPO, Dueling/Double DQN, RLHF, CEM planning, world models (PyBullet)
ML:	PyTorch, scikit-learn, sharpness-aware minimization, long-tailed recognition
Tools:	Transformers, LangChain, FAISS, Pandas, NumPy, Git, Linux, Docker, LaTeX